Customer Stories / Software & Internet / United Kingdom

2023

AI Startup Humanising Autonomy Scales Machine Learning Services Using AWS

Humanising Autonomy built new digital AI services using AWS, helping it unlock future scalability, deliver rapid processing performance, and maintain its own strict privacy and ethics rules. Humanising Autonomy, an AI startup founded in 2017, was already using Amazon EC2 as an AI training ground—but it needed new capabilities to help it scale up and expand its customer base. The solution uses multiple AWS services to give Humanising Autonomy a powerful basis for future growth, helping it process videos faster and distribute its software to customers.

Overview | **Opportunity** | **Solution** | **Outcome** | **AWS Services Used**

Supports

scalable growth through AI services

94%

Customer data

security improved

Streamlined delivery

of software using APIs

Overview

Humanising Autonomy was already using Amazon Web Services (AWS) for training, researching, and refining its machine learning (ML) models. But as the business grew, it needed a more scalable and flexible way to deliver software to help boost its engineering capabilities, such as video processing. Humanising Autonomy decided to expand its use of AWS and now uses multiple services for its customer-facing website, which lets users access software and other services. The company processes a vast amount of data uploaded from CCTV cameras, in-vehicle dashcams, and other input devices. Whether it's metadata or raw data generated from these devices, Humanising Autonomy uses AWS encryption services—such as AWS Key Management Service and AWS CloudHSM—to take measures to protect, secure, and anonymize data that it processes on behalf of its customers. This means that Humanising Autonomy needed robust technical and organizational measures to ensure its compliance with data privacy laws that apply to the use of these categories of data. In early 2022 Humanising Autonomy started to scale out its existing AWS architecture and build new services. It maintained its local edge machines for near real-time image processing, but used AWS for many of its workloads, from finance and human resources systems to sandboxed AI development and research, and delivering final software to customers.

Opportunity | Humanising Autonomy Seizes the Chance to Grow its Existing Services

Humanising Autonomy, founded in 2017 by Maya Pindeus, Raunaq Bose, and Leslie Nooteboom, develops ethical behavior AI that is changing how machines understand people. The company's goal is to make automation human centric by teaching machines to better understand people and using this layer of human context to help companies develop next-generation products and services, create safer environments for people, and to elevate the customer experience. Humanising Autonomy's AI software analyzes, classifies, and interprets human behavior captured from video cameras, using both live, real-time footage, and historic, pre-recorded footage.

Humanising Autonomy's technology considers the cognitive processes people use every day to understand the social and environmental context of what's happening around them—and translates this into machines can understand. For example, in Nextbase dashcams, Humanising Autonomy's softw n warnings and driver blind spots. For Transport for London, Humanising Autonomy's software s improve safety through automated emergency brakes, front and near-side driver alerts, or rear mirror camera monitoring systems.

Humanising Autonomy built its original infrastructure on Amazon Elastic Compute Cloud (Amazon EC2), which offers secure and resizable compute capacity for virtually any workload. This was the foundation for its research-based services, which allowed engineers to experiment and test new AI models.

Humanising Autonomy needed to expand its infrastructure to support the growth of the company, to scale its business, and to deliver a more efficient, effective, and professional software distribution service to its customers. “We’ve always had a combined cloud and edge-based technology stack,” says Raunaq Bose, co-founder and chief executive officer (CEO) of Humanising Autonomy. “We realized that by expanding our existing use of AWS we could scale faster and deliver a better customer experience.”

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This means that Humanising Autonomy needed robust technical and organizational measures to ensure its compliance with data privacy laws that apply to the use of these categories of data. “Data privacy compliance and ethical use of AI are central to our business model. We needed a cloud infrastructure that followed AWS Well-Architected principles and was designed for our ethical AI policies from the ground up,” says Bose. “To ensure the privacy of our customers’ data, we didn’t want a connection between our customers’ data and our systems.”

Solution | Improving Performance and Safeguarding Privacy Using AWS Services

In early 2022 Humanising Autonomy started to scale out its existing AWS architecture and build new services. It maintained its local edge machines for near real-time image processing, but used AWS for many of its workloads, from finance and human resources systems to sandboxed AI development and research, and delivering final software to customers.

Under the previous system, customers were emailed the access details for the software or API keys. With its new services, Humanising Autonomy offers its customers a much smoother solution: an online portal that acts as the gateway to its services and software. This capability is made possible by Amazon Simple Storage Service (Amazon S3), which is capable of retrieving vast amounts of data from multiple locations to allow Humanising Autonomy's customers to connect to its software.

The company can now also distribute its software, software development kit, and API keys to customers more efficiently using Amazon API Gateway, which creates, maintains, and secures APIs at any scale. This gives Humanising Autonomy's customers several options when it comes to running the software. They can either host the software locally on their own servers, run it on their own cloud service, or run it from Humanising Autonomy's own AWS servers. "Amazon API Gateway lets us deliver a slicker experience to customers, giving them access to our software remotely," says Bose.

Humanising Autonomy uses a number of services to uphold applicable data privacy standards and the company's policies on the ethical use of personal data (for example, any persons appearing in images taken by any of the cameras or input devices owned by its customers). The services—AWS X-Ray, Amazon CloudWatch, and AWS Control Tower—help Humanising Autonomy monitor and manage internal developer access to such personal data, ensure tight controls over its personnel to maintain high compliance and security standards, and deliver appropriate separation firewalls over its customer's data.

Humanising Autonomy has used AWS to build a serverless online platform and a scalable infrastructure that can expand as the business grows. Key parts of the solution include AWS Batch, which lets customers run batch processing, ML model training, and analysis at any scale, and Amazon EventBridge, which helps build event-driven applications at scale across AWS, existing systems, or SaaS applications. These services work together to automate large batch operations, such as video processing.

Using AWS Lambda, which lets customers run code without thinking about servers or clusters, Humanising Autonomy has been able to take advantage of serverless technology. The serverless architecture means Humanising Autonomy pays only for what it uses, keeping costs to a minimum. Overall, its use of AWS services means it doesn't have to reinvent the wheel. "We now have a whole suite of capabilities that we don't have to build ourselves," says Bose. "AWS technologies do the hard work for us, and the infrastructure manages itself."

Compute performance is crucial for AI and other services such as computer vision. The AWS infrastructure means near real-time and historic, non-real-time videos can now be processed in minutes. "It used to take us a day to process around 100 videos, with an engineer needed on-hand to oversee the process," says Bose. "Using AWS, it takes just 30 minutes and all it needs is a few minutes to set up. The engineer is then free to work on other tasks. It's a significant time-saving compared to the previous video-processing system."

Humanising Autonomy has also joined AWS Marketplace as an Amazon Partner Network-listed supplier. This will mean the company can co-sell with AWS via the marketplace and the APN Customer Engagements (ACE) program to reach new audiences across multiple industries.

Outcome | Delivering a Robust Platform, Better Performance, and Ethical Data Handling

Using AWS, Humanising Autonomy has launched its new software-as-a-service platform to im ers' experience, make engineering processes smoother, and help it grow. "We couldn't have done t ," says Bose. "It's a whole new infrastructure for us—it lets us distribute our product more effect performance, and gives us the flexibility we need to grow. Our software becomes more accessible to our customers, and that's really important. It's frictionless and easy to use."

About Humanising Autonomy

Humanising Autonomy's ethical AI technology—used in dashcams, car safety systems, smart cities, and more—extracts actionable insights from video data based on human behavior. The company's mission is to use AI to change how machines understand people.

AWS Services Used

Amazon S3

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.

Learn more »

Amazon EventBridge

Build event-driven applications at scale across AWS, existing systems, or SaaS applications.

Learn more »

Amazon CloudWatch

Amazon CloudWatch is a service that monitors applications, responds to performance changes, optimizes resource use, and provides insights into operational health.

Learn more »

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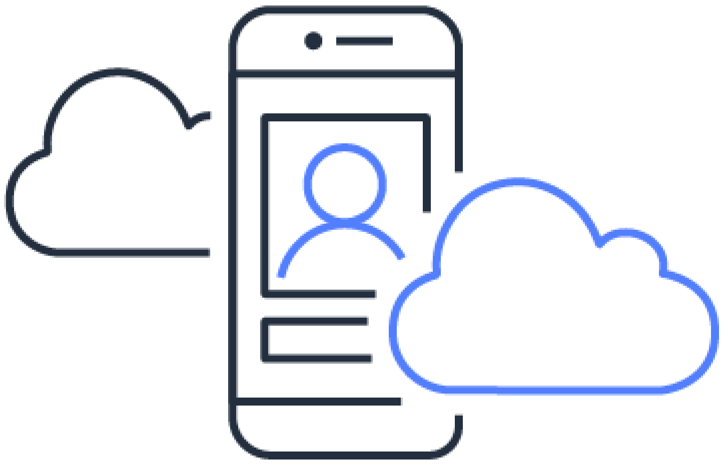

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